TASK 5 – Header Files and Conditional Compilation

5.1 Example project structure:

```
billing_project/
  main.c
  tariff.c
  billing.h
  README.md
```

5.2 Header file: billing.h:

```
#ifndef BILLING_H
#define BILLING_H

#define SLAB1_UNITS 100
#define SLAB2_UNITS 200
#define RATE1_PAISE 0
#define RATE2_PAISE 225
#define RATE3_PAISE 450
#define FIXED_CHARGE_PAISE 5000LL

long long compute_bill(int units);
unsigned int consumption(unsigned short previous,
                         unsigned short current);
#endif
```

5.3 Implementation file: tariff.c:

```
#include "billing.h"
#include <stdio.h>

unsigned int consumption(unsigned short previous,
                         unsigned short current)
{
    return (unsigned short)(current - previous);
}
long long compute_bill(int units)
{
    long long paise = FIXED_CHARGE_PAISE;
#ifdef DEBUG
    fprintf(stderr, "[DEBUG] units=%d, fixed=%lld paise\n",
            units, paise);
#endif
    if (units > SLAB2_UNITS) {
        paise += (SLAB2_UNITS - SLAB1_UNITS) *
RATE2_PAISE;
        paise += (long long)(units - SLAB2_UNITS) *
RATE3_PAISE;
    } else if (units > SLAB1_UNITS) {
        paise += (long long)(units - SLAB1_UNITS) *
RATE2_PAISE;
    }
    return paise;}
```

5.4 Main file: main.c:

```
#include <stdio.h>
#include "billing.h"

int main(void)
{
    int units = 150;
    long long paise = compute_bill(units);
    printf("%d units -> Rs %lld.%02lld\n",
           units, paise / 100, paise % 100);
    return 0;
}
```

Compile normally with gcc -Wall -Wextra -std=c11 main.c tariff.c -o billing. Compile with debug output enabled using gcc -Wall -Wextra -std=c11 -DDEBUG main.c tariff.c -o billing_debug. #ifdef DEBUG includes the debugging statements only when DEBUG is defined at compilation.

Include guards prevent a header from being processed repeatedly within one translation unit. #include <file> typically searches the implementation's system include paths; #include "file" typically searches the current/source directory first, then configured include paths.

5.5 Compile-time versus run-time debug flag:

With #ifdef DEBUG, a non-debug build has no debug statement in the compiled program and no run-time condition to check. With if (debug), the executable generally retains the debug code and each guarded location performs a condition test. The run-time approach allows debug mode to be switched while the program runs; the preprocessor approach creates separate debug and normal builds. The additional test is O(1) per guarded call.